DHCP Practical Recreation Instruction Manual

SIT716 HD Plus Project - DHCP Normal DORA and Rogue DHCP Evidence Reproduction

This manual explains how another user can recreate the DHCP practical environment, import the Ubuntu OVA into VirtualBox, run the DHCP normal and rogue DHCP scenarios, capture the traffic in Wireshark, save evidence, and compare the results with the website simulation.

Project area	DHCP protocol simulation
Main goal	Reproduce normal DHCP DORA and rogue DHCP offer behaviour
Main tools	Oracle VirtualBox, Ubuntu VM, MiniEdit/Mininet, Wireshark, terminal
Key evidence	DHCP Discover, Offer, Request, ACK, and Rogue Offer
Website file	dhcp.html with dhcp.js simulation controls

1. Required Files and Software

Before starting, download the DHCP reproduction artefacts from the shared project folder. Keep the OVA, PCAP files, screenshots, and this PDF in one local folder so the evidence remains organised.

- Oracle VirtualBox installed on the host computer.
- DHCP Ubuntu OVA file downloaded from the project folder.
- DHCP PCAP files downloaded for review in Wireshark.
- At least 4 GB RAM available for the VM; 6 GB or more is recommended.
- A stable local folder path, for example D:\Desktop\SIT716-DHCP-Recreation.

2. Import the Ubuntu OVA in VirtualBox

The OVA contains the prepared Ubuntu environment used to recreate the DHCP practical. Import it before running any DHCP tests.

- Open Oracle VirtualBox.
- Click File -> Import Appliance.
- Browse to the downloaded DHCP OVA file.
- Keep the default virtual hardware settings unless your computer has limited RAM.
- Click Finish or Import and wait for the VM to be created.
- Start the imported Ubuntu VM and log in using the provided VM credentials from the project folder or classroom instructions.

Verification checkpoint: confirm that Ubuntu opens successfully and that the left dock, terminal, MiniEdit/Mininet, and Wireshark are available.

3. Prepare the Ubuntu Lab Environment

Open a terminal inside Ubuntu and verify the tools. The exact installed packages may already exist in the OVA. These commands are useful for checking the environment.

```
which wireshark
which mn
python3 --version
sudo mn --version
ip addr
```

If Mininet is already installed, continue. If a package is missing and the VM has internet access, install only the missing dependency.

```
sudo apt update
sudo apt install -y mininet wireshark dnsmasq isc-dhcp-client
```

4. Open or Build the MiniEdit Topology

Use the topology supplied in the OVA or project folder. If the topology file is available, open it in MiniEdit. The expected DHCP lab uses a simple switched network with a client, legitimate DHCP server, and attacker or rogue DHCP host.

```
sudo python3 /usr/bin/miniedit.py
```

If the OVA uses a different MiniEdit path, locate it with:

```
sudo find /usr -iname "miniedit.py" 2>/dev/null
```

Recommended logical host roles for the DHCP test:

- Client host: requests a DHCP lease.
- Legitimate DHCP server: provides the trusted address lease.
- Rogue DHCP attacker: sends an unauthorised DHCP offer.
- Switch: connects all hosts in the same broadcast domain for controlled testing.

Example addressing used in the evidence screenshots can include a legitimate server around 192.168.60.50, a client lease such as 192.168.60.101, and a rogue responder such as 192.168.60.219. Use the exact addressing supplied in your OVA or lab notes if it differs.

5. Start Wireshark Capture

Start Wireshark before running the DHCP process so the complete Discover, Offer, Request, and ACK sequence is captured.

- Open Wireshark from the Ubuntu dock or terminal.
- Select the interface connected to the Mininet switch, commonly s1-eth1 or another s1 interface.
- Use this display filter: bootp or dhcp or arp or icmp.
- Start capture before requesting a DHCP lease.

```
bootp or dhcp or arp or icmp
```

Note: Wireshark may label DHCP as BOOTP because DHCP is based on BOOTP packet structure.

6. Run Normal DHCP DORA Test

The normal test demonstrates a trusted DHCP server assigning configuration to a client. Use Mininet xterms so each host can run its own command.

```
# In Mininet CLI, open terminals for client and server
xterm h1 h2

# Example role mapping:
# h1 = client
# h2 = legitimate DHCP server
```

On the legitimate DHCP server terminal, create or use a DHCP configuration. The following dnsmasq example is suitable for a controlled lab. Adjust interface names and IP ranges to match your topology.

```
# On legitimate DHCP server h2
sudo ip addr flush dev h2-eth0
sudo ip addr add 192.168.60.50/24 dev h2-eth0
sudo ip link set h2-eth0 up

sudo dnsmasq --no-daemon --interface=h2-eth0 --bind-interfaces
--dhcp-range=192.168.60.100,192.168.60.120,255.255.255.0,1h --dhcp-option=3,192.168.60.1
--dhcp-option=6,192.168.60.50 --log-dhcp
```

On the client terminal, clear any existing IP address and request a new DHCP lease.

```
# On client h1
sudo ip addr flush dev h1-eth0
```

```
sudo dhclient -r h1-eth0
sudo dhclient -v h1-eth0
ip addr show h1-eth0
ip route
```

Expected Wireshark evidence: DHCP Discover from the client, DHCP Offer from the legitimate server, DHCP Request from the client, and DHCP ACK from the legitimate server.

7. Run Rogue DHCP Attack Test

The rogue DHCP test demonstrates how an attacker can compete with the legitimate DHCP server and provide unauthorised configuration. Use this only inside the isolated lab VM/topology.

```
# In Mininet CLI, open a terminal for the attacker
xterm h3

# Example role mapping:
# h3 = rogue DHCP attacker
```

On the attacker host, start a rogue DHCP service. Adjust the interface and values to match the lab topology.

```
# On attacker h3
sudo ip addr flush dev h3-eth0
sudo ip addr add 192.168.60.219/24 dev h3-eth0
sudo ip link set h3-eth0 up

sudo dnsmasq --no-daemon --interface=h3-eth0 --bind-interfaces
--dhcp-range=192.168.60.200,192.168.60.230,255.255.255.0,1h --dhcp-option=3,192.168.60.219
--dhcp-option=6,192.168.60.219 --log-dhcp
```

On the client, release the previous lease and request a new one while Wireshark is capturing.

```
# On client h1
sudo dhclient -r h1-eth0
sudo ip addr flush dev h1-eth0
sudo dhclient -v h1-eth0
ip addr show h1-eth0
ip route
```

Expected Wireshark evidence: the rogue DHCP host may send a competing DHCP Offer. Compare the server identifier, offered IP address, router option, and DNS option to determine whether the offer is trusted or malicious.

8. Save PCAP Evidence

After capturing the normal and rogue DHCP tests, save the traffic as PCAP or PCAPNG so it can be used in the website evidence and final demonstration.

- In Wireshark, click File -> Save As.
- Save the normal capture as dhcp-normal-dora.pcapng.
- Save the rogue capture as dhcp-rogue-offer.pcapng.
- Keep screenshots of the packet list and packet details pane for the website evidence section.

9. Website Comparison Checklist

After completing the lab, return to the website DHCP page and compare the practical evidence with the simulation.

- Run the predefined normal DHCP demo and compare it with the normal DORA PCAP.
- Run the predefined attack demo and compare it with the rogue DHCP offer PCAP.
- Confirm the website event log explains Discover, Offer, Request, ACK, and Rogue Offer clearly.
- Use the manual validation table to test valid HQ and BR DHCP paths plus wrong-branch invalid examples.
- Record a screenshot of the website simulation and one Wireshark evidence screenshot for the final demonstration.

10. Troubleshooting

Problem	Likely cause	Fix
No DHCP packets appear	Capture started on wrong interface	Try s1-eth1, any, or the host interface shown in Mininet.
Client keeps old IP	Previous lease still active	Run dhclient -r and flush the interface before requesting again.
dnsmasq fails to start	Interface already in use or wrong interface name	Check ip link and stop previous dnsmasq process.
Only legitimate offer appears	Rogue server not started or too slow	Start rogue dnsmasq before client requests a lease.
Website test button disabled	Invalid source/destination pair	Use the IP reference table in dhcp.html and validate again.

Safety Note

Run rogue DHCP demonstrations only inside the isolated VirtualBox and Mininet lab. Do not run rogue DHCP services on a real campus, workplace, home, or shared network because it can disrupt other devices by giving them incorrect network settings.